

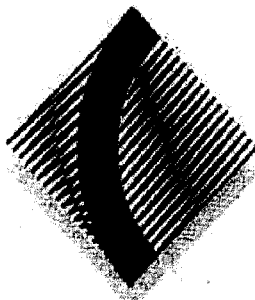
AT
KW
KL
JG

Name: _____

Class: 12MTX _____

Teacher: _____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2010 AP4

YEAR 12 TRIAL HSC EXAMINATION

MATHEMATICS EXTENSION 1

*Time allowed - 2 HOURS
(Plus 5 minutes' reading time)*

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- All questions are of equal value.
- Each question is to be commenced on a new page clearly marked Question 1, Question 2, etc on the top of the page. **
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used. Standard Integral Tables are provided
- Your solutions will be collected in one bundle stapled in the top left corner.

Please arrange them in order, Q1 to 7.

*****Each page must show your name and your class. *****

Question 1 (12 Marks)**Marks**

- (a) Two points A and B have coordinates $(-2,4)$ and $(2,1)$ respectively.
Find the point P which divides the interval AB externally in the ratio $3 : 2$. **2**
- (b) Find the acute angle between the lines $y=3x-1$ and $4x-2y=7$ **2**
- (c) Solve the inequality $\frac{2x-1}{x+1} \geq 1$. **3**
- (d) The equation $x^3 - 2x^2 + 4x - 5 = 0$ has roots α, β, γ
- (i) Write down the value of $\alpha\beta + \beta\gamma + \gamma\alpha$ **1**
- (ii) Find the value of $\alpha\beta\gamma$ **1**
- (ii) Hence find the value of $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ **2**
- (e) Find $\lim_{\theta \rightarrow 0} \frac{\tan 2\theta}{8\theta}$ **1**

End of Question 1.**Question 2 (12 Marks) Start a new page**

- (a) $P (2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$.
- (i) Show that the equation of the normal at P is **2**
- $$x + py = 2ap + ap^3$$
- (ii) Q is the point with coordinates $(0, 2a + ap^2)$, find the coordinates of M , the midpoint of PQ . **1**
- (iii) Find the equation of the locus of M , **1**

Question 2 continued.....**Marks**

(b) Use the substitution $u = 1 - x$ to evaluate $\int_{-15}^0 \frac{x \, dx}{\sqrt{1-x}}$.

3

(c) Find $\int \sin^2 6x \, dx$

2

(d) (i) Sketch the graph $y = |x - 2|$

1

(ii) Hence solve for x

$$|x - 2| < \frac{1}{2}x$$

2**End of Question 2.****Question 3 (12 Marks) Start a new page**

(a) Solve the equation $\sin 2x + \cos x = 0$ for $0 \leq x \leq 2\pi$.

3

(b) (i) Use the substitution $t = \tan \frac{\theta}{2}$ to show that

$$\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}$$

2

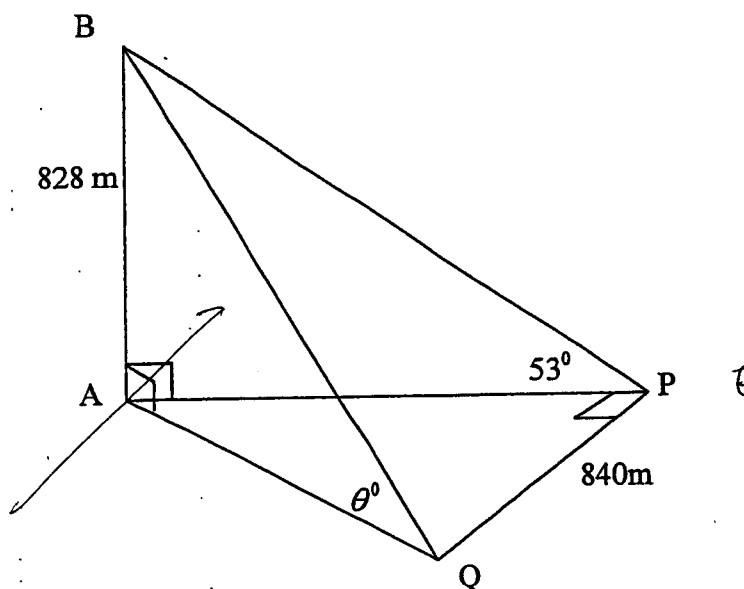
(ii) Hence show that

3

 $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\operatorname{cosec} \theta + \cot \theta) \, d\theta = \ln 2$

- (c) A tower is 828 metres tall. At a point due east of the tower, the angle of elevation is 53° . At another point due south of P , the angle of elevation is θ degrees. The distance from P to Q is 840 metres.

NOT TO SCALE



(i) Prove that $\cot \theta = \sqrt{\frac{828^2 \cot^2 53^\circ + 840^2}{828^2}}$

3

- (ii) Find θ , correct to the nearest degree.

1

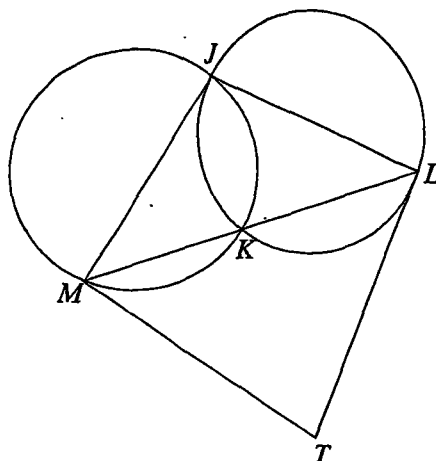
End of Question 3.

Question 4 (12 Marks) Start a new page

Marks

- (a) The circles intersect at J and K , LKM is a straight line.
 TL and TM are tangents. Let $\angle LMT = \alpha$ and $\angle MLT = \beta$.

NOT TO SCALE



Copy the diagram onto your answer sheet

Hence prove that $TMJL$ is a cyclic quadrilateral.

3

- (b) A particle is moving along a straight line with acceleration $\ddot{x} = 8x^3$,
 where x is the displacement from the origin O in metres. Initially the particle
 is 1 metre to the right of the origin moving with velocity $v = 2 \text{ ms}^{-1}$.

(i) Show that $\ddot{x} = \frac{d(\frac{1}{2}v^2)}{dx}$.

1

(ii) Show that $v^2 = 4x^4$.

2

- (iii) Explain why the velocity v cannot be negative.

1

- (iv) Find the time taken for the particle to travel to a point 2 metres
 to the right of the origin.

2

- (c) Let T be the temperature of a cup of tea at time t minutes after it has been brought into a room of temperature A . The Newton's Law of Cooling states that the rate of change of temperature T is proportional to $(T - A)$,

that is $\frac{dT}{dt} = -k(T - A)$, where k is a positive constant.

- (i) Show that $T = A + Be^{-kt}$, where B is a constant, satisfies Newton's Law of Cooling. 1

- (ii) A cup of tea, with an initial temperature of 95°C , is brought into a room of temperature 25°C . After 10 minutes, the temperature of the tea drops to 60°C . Find the temperature of the tea after another 5 minutes. 2

End of Question 4.

Question 5 (12 Marks) Start a new page

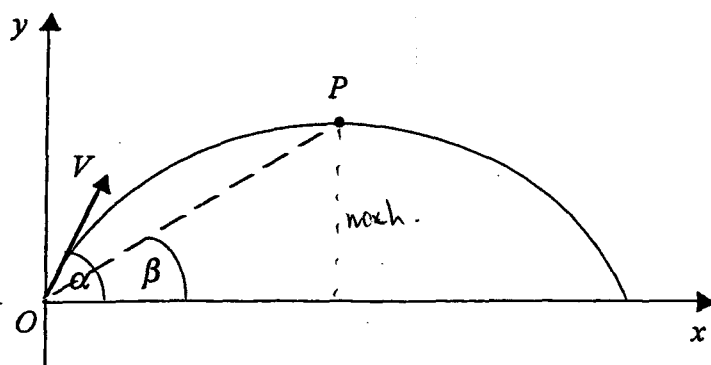
- (a) A particle is moving in a straight line so that its displacement x metres from a fixed point on the line at any time t seconds is given by

$$x = \frac{3}{2} \sin 2t + 2 \cos 2t$$

- (i) Show that $\ddot{x} = -n^2x$, describe the motion of the particle and state the period of the motion 2
- (ii) Find the maximum displacement of the particle. 2

(b)

NOT TO SCALE



A particle is projected from a point O with speed $V \text{ ms}^{-1}$ at an angle α radians above the horizontal, where $0 \leq \alpha \leq \frac{\pi}{2}$. It moves in a vertical plane subject to gravity where the acceleration due to gravity is 10 ms^{-2} . At time t seconds, it has horizontal and vertical displacements x metres and y metres respectively from O . At point P where it attains its greatest height the angle of elevation of the particle from O is β radians.

(i) Use integration to show that $x = Vt \cos \alpha$ and $y = Vt \sin \alpha - 5t^2$ 2

(ii) Show that $\tan \beta = \frac{1}{2} \tan \alpha$. 3

(iii) If the particle has greatest height 80 metres above O at a horizontal distance 120 metres from O , find the exact values of α and V . 3

End of Question 5.

Question 6 (12 Marks) Start a new page**Marks**

- (a) Find $\frac{d}{dx}(e^x \tan^{-1} x)$ **2**
- (b) Find the exact value of $\int_1^{\sqrt{3}} \frac{2 dx}{\sqrt{4-x^2}}$ **3**
- (c) (i) State the domain and range for $y = \cos^{-1} 4x$ **1**
- (ii) Sketch the graph of $y = \cos^{-1} 4x$ **1**
- (iii) Find the equation of the tangent to $y = \cos^{-1} 4x$ at $x = -\frac{1}{8}$.
—/— Leave answer in exact form. **3**
- (d) (i) Sketch the graphs of $y = \sin^{-1} x$, for $0 \leq x \leq 1$
and $y = \sin x$, for $0 \leq x \leq \frac{\pi}{2}$ on separate diagrams. **1**
- (ii) By considering your graphs drawn in part (i), find the exact value of
 $\int_0^1 \sin^{-1} x \, dx + \int_0^{\frac{\pi}{2}} \sin x \, dx$ **1**

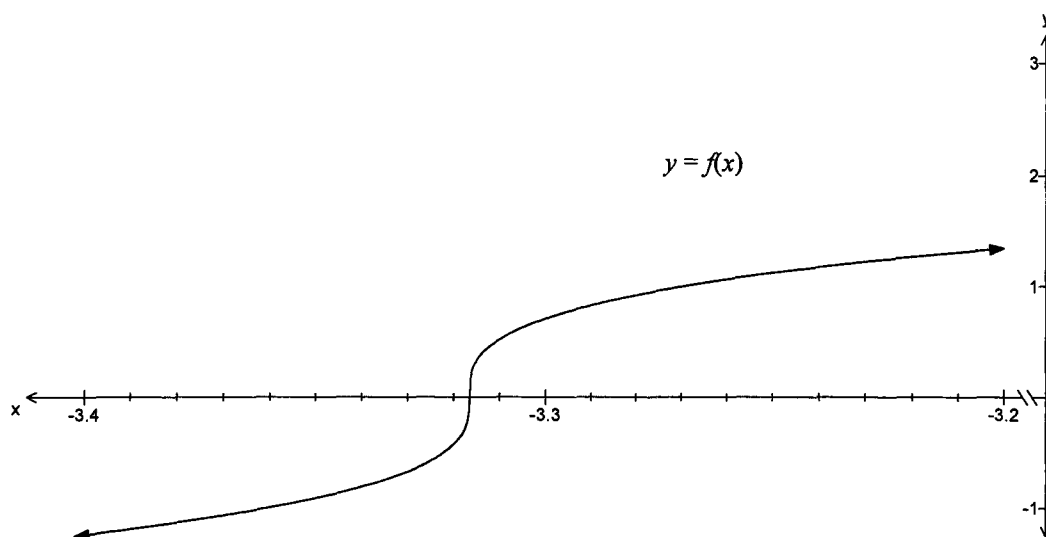
End of Question 6.**Question 7 (12 Marks) Start a new page**

- (a) Prove, using the method of mathematical induction,
that $4^n + 14$ is divisible by 6 for all $n \geq 1$. **3**

Question 7 continued.....

Marks

- (b) The curve $f(x) = (x^3 - 12x)^{\frac{1}{3}}$ is shown below.



- (i) Find $f'(x)$. (No need to simplify your answer.)

1

- (ii) Taking an initial estimate of $x_1 = -3.3$,

use one application of Newton's Method to obtain another approximation to the root of $f(x) = 0$.

1

- (iii) Explain why using $x_1 = -3.3$ does not produce a better approximation to the root than the original estimate.

1

- (c) Find the term independent of x in the expansion of $\left(x^2 - \frac{2}{x}\right)^9$

2

Question 7 Continued.....**Marks**

- (d) (i) Using the expansion of $(1+x)^{n-1}$ show that

$$\binom{n-1}{1} + \binom{n-1}{2} + \dots + \binom{n-1}{n-2} = 2^{n-1} - 2.$$

2

- (ii) Find the least positive integer n , such that

$$\binom{n-1}{1} + \binom{n-1}{2} + \dots + \binom{n-1}{n-2} > 1\,000$$

2**END OF EXAM**

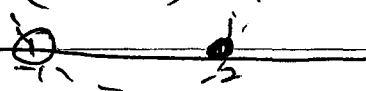
Ext 1 TRIAL 2010

a) $3! - 2$
 $P = \left(\frac{3 \times 2 + -2 \times -2}{3 + (-2)}, \frac{3 \times 1 + -2 \times 4}{3 + (-2)} \right)$
 $= (10, -5)$

b) $y = 3x - 1$ $4x - 2y = 7$
 $\therefore m_1 = 3$ $2y = 4x - 7$
 $y = 2x - \frac{7}{2}$
 $\therefore m_2 = 2$

$\tan \theta = \left| \frac{3 - 2}{1 + 3 \times 2} \right|$
 $= \frac{1}{7}$
 $\theta = 8^\circ 8'$

c) $(x+1)(2x-1) \geq (x+1)^2$
 $(x+1)(2x-1) - (x+1)^2 \geq 1$
 $(x+1)(2x-1-x-1) \geq 1$
 $(x+1)(x-2) \geq 1$

$x \neq -1$ 
 $x < -1$ or $x \geq 2$

d) (i) $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$
 $= 4$

(ii) $\alpha\beta\gamma = -\frac{d}{a}$
 $= 5$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

$\frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = \frac{4}{5}$

e) $\lim_{\theta \rightarrow 0} \frac{\tan 2\theta}{2\theta} = \frac{1}{2}$

Q2 a) $y = \frac{x^2}{2a}$

(i) $\frac{dy}{dx} = \frac{x}{a}$

at $x = 2ap$

$\frac{dy}{dx} = p$

$\therefore$ grad of normal $= -\frac{1}{p}$

Equation of normal

$y - ap^2 = -\frac{1}{p}(x - 2ap)$

$py - ap^3 = -x + 2ap$

$x + py = 2ap + ap^3$

(ii) $M = \left(\frac{2ap}{2}, \frac{ap^2 + 2a + ap^2}{2} \right)$
 $= (ap, a + ap^2)$

(iii) $x = ap$

$p = \frac{x}{a}$

$y = a(1 + p^2)$
 $= a\left(1 + \frac{x^2}{a^2}\right)$

$y = a + \frac{x^2}{a}$

$x^2 = ay - a^2$

$x^2 = a(y - a)$

Q2

b) $u = 1 - x$
 $du = -dx$

When $x = -15$, $u = 16$

$x = 0$, $u = 1$

$$\int_{-15}^0 \frac{x dx}{\sqrt{1-x}} = - \int_{16}^1 \frac{1-u}{\sqrt{u}} du$$

$$= \int_{16}^1 u^{-\frac{1}{2}} - u^{\frac{1}{2}} du$$

$$= \left[2 \cdot u^{\frac{1}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right]_{16}^1$$

$$= \left(8 - \frac{128}{3} \right) - \left(2 - \frac{2}{3} \right)$$

$$= -36.$$

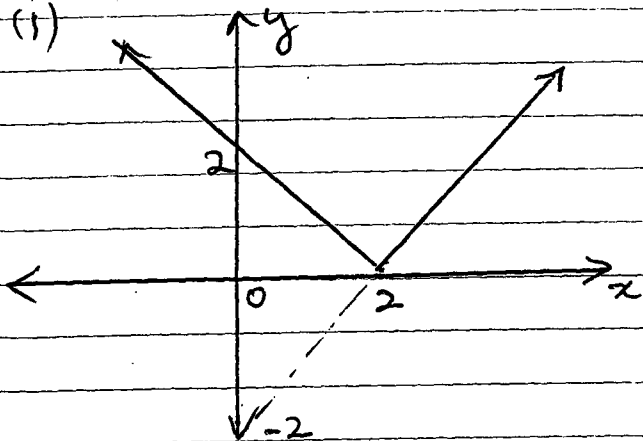
c) $\int \sin^2 6x dx$

$$= \frac{1}{2} \int (1 - \cos 12x) dx$$

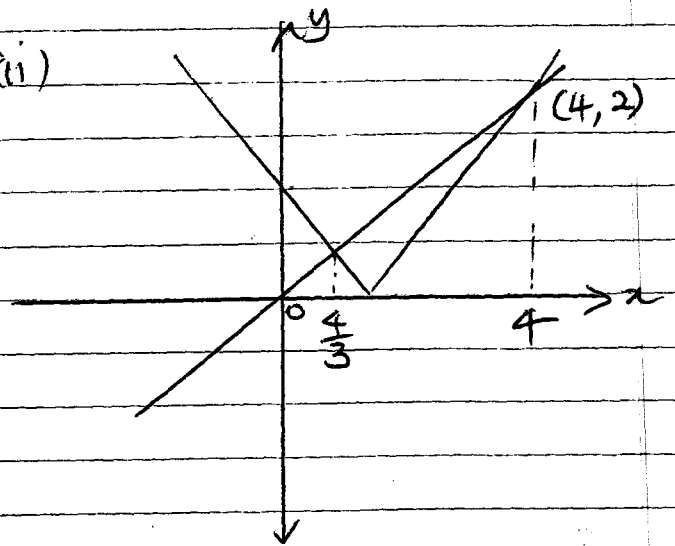
$$= \frac{1}{2} \left[x - \frac{\sin 2x}{12} \right] + C$$

$$= \frac{x}{2} - \frac{\sin 2x}{24} + C$$

d) (i)



(ii)



$$x - 2 = \frac{1}{2}x$$

$$\frac{1}{2}x = 2$$

$$x = 4$$

$$-x + 2 = \frac{1}{2}x$$

$$\frac{3}{2}x = 2$$

$$x = \frac{4}{3}$$

$$\therefore \frac{4}{3} < x < 4$$

Q3

$$(a) \sin 2x + \cos x = 0$$

$$2\sin x \cos x + \cos x = 0$$

$$\cos x (2\sin x + 1) = 0$$

$$\cos x = 0 \quad \text{or} \quad \sin x = -\frac{1}{2}$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2} \quad x = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$b) (i) t = \tan \frac{\theta}{2}$$

$$\operatorname{cosec} \theta = \frac{1+t^2}{2t}$$

$$\cot \theta = \frac{1-t^2}{2t}$$

$$\begin{aligned} \text{LHS} &= \operatorname{cosec} \theta + \cot \theta \\ &= \frac{1+t^2}{2t} + \frac{1-t^2}{2t} \end{aligned}$$

$$= \frac{2}{2t}$$

$$= \frac{1}{t}$$

$$= \cot \frac{\theta}{2} = \text{RHS}$$

$$(ii) \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\operatorname{cosec} x + \cot x) dx$$

$$= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot \frac{x}{2} dx$$

$$= 2 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\frac{1}{2} \cos \frac{x}{2}}{\sin \frac{x}{2}} dx$$

$$= 2 \left[\ln \left(\sin \frac{x}{2} \right) \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= 2 \left[\ln \left(\sin \frac{\pi}{4} \right) - \ln \left(\sin \frac{\pi}{6} \right) \right]$$

$$= 2 \left[\ln \frac{1}{\sqrt{2}} - \ln \frac{1}{2} \right]$$

$$= 2 \ln (2^{-\frac{1}{2}} \div 2^{-1})$$

$$= 2 \ln 2^{\frac{1}{2}}$$

$$= \ln 2$$

$$(i) \text{ In } \triangle PAB$$

$$\tan 53^\circ = \frac{828}{AP}$$

$$AP = 828 \cot 53^\circ$$

$$\text{In } \triangle ABQ$$

$$\tan \theta = \frac{828}{AQ}$$

$$AQ = 828 \cot \theta$$

$$\text{In } \triangle APQ$$

$$AP^2 + 840^2 = AQ^2$$

$$828^2 \cot^2 53^\circ + 840^2 = 828^2 \cot^2 \theta$$

$$\cot^2 \theta = \frac{828^2 \cot^2 53^\circ + 840^2}{828^2}$$

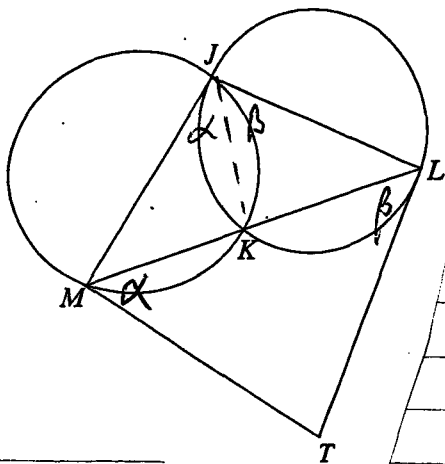
$$\cot \theta = \sqrt{\frac{828^2 \cot^2 53^\circ + 840^2}{828^2}}$$

$$(ii) \theta = 38.35^\circ$$

$$= 38^\circ$$

Q4

a)



$$\angle TMR = \angle MJK$$

$= \alpha$ (alternate segment theorem)

$$\angle K LJ = \angle K JL$$

$= \beta$ (alternate segment theorem)

In ΔLMT

$$\angle LTM = 180 - (\alpha + \beta) \text{ (sum of } \Delta)$$

$$\angle MJL = \alpha + \beta \text{ (adjacent } \angle\text{'s)}$$

$$\therefore \angle MJL + \angle LTM$$

$$= (\alpha + \beta) + 180 - (\alpha + \beta)$$

$$= 180^\circ$$

$\therefore$ opposite $\angle$'s supplementary

$\therefore TMJL$ is cyclic quadrilateral

$$\Rightarrow \text{ii) } \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \cdot \frac{dv}{dx}$$

$$= v \frac{dv}{dx}$$

$$= \frac{dx}{dt} \cdot \frac{dv}{dx}$$

$$= \frac{dv}{dt}$$

$$= \ddot{x}$$

OR

$$\ddot{x} = \frac{dv}{dt}$$

$$= \frac{dx}{dt} \times \frac{dv}{dx}$$

$$= v \times \frac{dv}{dx}$$

$$= \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \frac{dv}{dx}$$

$$= \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$\text{ii) } \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 8x^3$$

$$\frac{1}{2} v^2 = \int 8x^3 dx$$

$$= 2x^4 + C$$

When $x=1$, $v=2$, $\therefore C=0$.

$$\frac{1}{2} v^2 = 2x^4$$

$$v^2 = 4x^4$$

iii) Since $x \geq 1$, $\ddot{x}$ accelerates $8x^3 > 0$, ~~velocity~~ $v^2 = 4x^4$ which ≥ 0 and initial $v > 0$ $\therefore v$ cannot be negative.

$$\text{iv) } v > 0 \quad \therefore v = 2x^2$$

$$\frac{dx}{dt} = 2x^2$$

$$\frac{dt}{dx} = \frac{1}{2x^2}$$

$$t = \frac{1}{2} \int x^{-2} dx$$

$$= -\frac{1}{2} x^{-1} + C$$

When $t=0$, $x=1$, $C = \frac{1}{2}$

$$t = -\frac{1}{2} x^{-1} + \frac{1}{2}$$

When $x=2$

$$t = -\frac{1}{2} \times 2^{-1} + \frac{1}{2} = \frac{1}{4} \text{ seconds}$$

$$\text{OR } t = \int_1^2 \frac{1}{2x^2} dx$$

$$= \frac{1}{2} [-x^{-1}]_1^2$$

$$= \frac{1}{2} \left[-\frac{1}{2} + 1 \right]$$

$$= \frac{1}{4} \text{ seconds}$$

$\checkmark$ a better answer.

Initially $v > 0$ and $x=1$. To change direction $v < 0$, but $v < 0$ when $x=0$ $\therefore$ impossible

$$\begin{aligned}
 \text{c) i) } T &= A + B e^{-kt} \\
 \frac{dT}{dt} &= B e^{-kt} \times -k \\
 &= -k B e^{-kt} \\
 &= -k(A + B e^{-kt} - A) \quad | \\
 &= -k(T - A)
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } A &= 25 \\
 \therefore T &= 25 + B e^{-kt} \\
 \text{when } t &= 0, T = 95 \\
 95 &= 25 + B \\
 \therefore B &= 70 \\
 T &= 25 + 70 e^{-kt} \\
 \text{when } t &= 10, T = 60 \\
 60 &= 25 + 70 e^{-10k} \\
 70 e^{-10k} &= 35 \\
 e^{-10k} &= \frac{1}{2} \\
 k &= -\frac{1}{10} \ln \frac{1}{2} \quad | \\
 T &= 25 + 70 e^{(\frac{1}{10} \ln \frac{1}{2}) t} \\
 \text{when } t &= 15 \quad (\frac{1}{10} \ln \frac{1}{2}) \times 15 \\
 T &= 25 + 70 e^{(\frac{1}{10} \ln \frac{1}{2}) \times 15} \\
 &= 49.75^\circ \text{C} \quad |
 \end{aligned}$$

Q5.

$$\begin{aligned}
 \text{a) } x &= \frac{3}{2} \sin 2t + 2 \cos 2t \\
 \dot{x} &= 3 \cos 2t - 4 \sin 2t \\
 \ddot{x} &= -6 \sin 2t - 8 \cos 2t \quad | \\
 &= -4 \left[\frac{3}{2} \sin 2t + 2 \cos 2t \right] \\
 &= -4x
 \end{aligned}$$

$\therefore$ Motion is SHM.

oscillating about origin /
period = π seconds.

OR.

$$\begin{aligned}
 x &= R \sin(2t + \alpha) \\
 R &= \sqrt{\left(\frac{3}{2}\right)^2 + 2^2} = \frac{5}{2} \\
 x &= \frac{5}{2} \sin(2t + \alpha) \\
 \dot{x} &= 2 \times \frac{5}{2} \cos(2t + \alpha) \\
 &= 5 \cos(2t + \alpha) \\
 \ddot{x} &= -10 \sin(2t + \alpha) \\
 &= -4 \left(\frac{5}{2} \sin(2t + \alpha) \right) \\
 &= -4x
 \end{aligned}$$

$\therefore$ Motion is SHM with
centre $x = 0$

$$\begin{aligned}
 \text{and period } T &= \frac{2\pi}{2} \\
 &= \pi.
 \end{aligned}$$

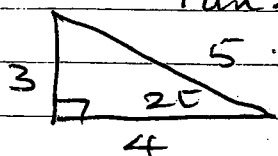
ii) Max displacement when $\dot{x} = 0$

$$3 \cos 2t - 4 \sin 2t = 0$$

$$4 \sin 2t = 3 \cos 2t$$

$$\frac{\sin 2t}{\cos 2t} = \frac{3}{4}$$

$$\tan 2t = \frac{3}{4} \quad |$$



$\therefore$ Max displacement

$$x = \frac{3}{2} \times \frac{3}{5} + 2 \times \frac{4}{5}$$

$$= 2.5 \text{ m} \quad |$$

OR

$$x = \frac{5}{2} \sin(2t + \alpha)$$

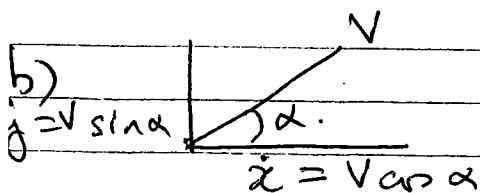
$$\text{Since } -1 \leq \sin(2t + \alpha) \leq 1$$

max displacement

$$\text{when } \sin(2t + \alpha) = 1$$

$$\therefore x = \frac{5}{2}$$

$$= 2.5 \text{ m}$$



$$x = \int V \cos \alpha \, dt$$

$$= Vt \cos \alpha + C_1$$

$$t=0, x=0 \therefore C_1 = 0$$

$$\therefore x = Vt \cos \alpha$$

$$\ddot{y} = -10$$

$$\dot{y} = -10t + C_2$$

$$t=0, \dot{y} = V \sin \alpha, C_2 = V \sin \alpha$$

$$\dot{y} = -10t + V \sin \alpha$$

$$y = -\frac{10t^2}{2} + Vt \sin \alpha + C_3$$

$$t=0, y=0, \therefore C_3 = 0$$

$$\therefore y = -5t^2 + Vt \sin \alpha$$

$$y = Vt \sin \alpha - 5t^2$$

ii) $\dot{y} = 0$ at greatest height

$$-10t + V \sin \alpha = 0$$

$$t = \frac{V \sin \alpha}{10}$$

Sub into eqn of x

$$x = \frac{V^2 \sin \alpha \cos \alpha}{10}$$

Sub t into eqn of y

$$y = -5 \frac{V^2 \sin^2 \alpha}{100} + \frac{V^2 \sin^2 \alpha}{10}$$

$$= \frac{5V^2 \sin^2 \alpha}{100}$$

$$\tan \beta = \frac{y}{x}$$

$$= \frac{5V^2 \sin^2 \alpha}{2 \times 100} \times \frac{10}{V^2 \sin \alpha \cos \alpha}$$

$$= \frac{1}{2} \frac{\sin \alpha}{\cos \alpha}$$

$$\therefore \tan \beta = \frac{1}{2} \tan \alpha$$

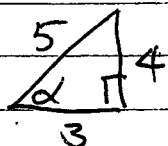
(ii) $P(120, 80)$

$$\tan \beta = \frac{80}{120} = \frac{2}{3}$$

from (i) $\tan \beta = \frac{1}{2} \tan \alpha$

$$\therefore \frac{1}{2} \tan \alpha = \frac{2}{3}$$

$$\tan \alpha = \frac{4}{3}$$



$$y = \frac{v^2 \sin^2 \alpha}{20}$$

$$80 = \frac{v^2 \times \left(\frac{4}{5}\right)^2}{20}$$

$$80 = \frac{1}{20} \times v^2 \times \frac{16}{25}$$

$$v^2 = 8 \times 20 \times \frac{25}{16}$$

$$= 2500$$

$$v = 50$$

Q6

(a) $\frac{d}{dx} [e^x \tan^{-1} x]$

$$= e^x \tan^{-1} x' + \frac{e^x}{1+x^2}$$

b) $\int_1^{\sqrt{3}} \frac{2}{\sqrt{4-x^2}} dx$

$$= \int_1^{\sqrt{3}} \frac{2}{\sqrt{2^2-x^2}} dx$$

$$= 2 \left[\sin^{-1} \frac{x}{2} \right]_1^{\sqrt{3}}$$

$$= 2 \left[\sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{1}{2} \right]$$

$$= 2 \left[\frac{\pi}{3} - \frac{\pi}{6} \right]$$

$$= 2 \times \frac{\pi}{6}$$

$$= \frac{\pi}{3}$$

c) i) Domain:

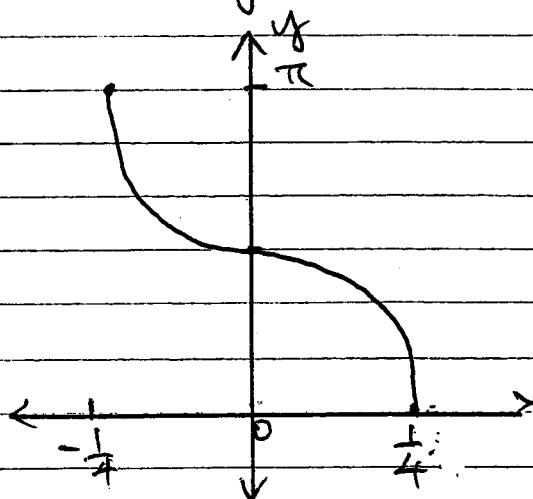
$$-1 \leq \sin x \leq 1$$

$$-\frac{1}{4} \leq x \leq \frac{1}{4}$$

Range:

$$0 \leq y \leq \pi$$

ii)



$$\text{ii) } y = \cos^{-1} 4x$$

$$\frac{dy}{dx} = \frac{-4}{\sqrt{1-16x^2}}$$

$$\text{at } x = -\frac{1}{8}$$

$$\frac{dy}{dx} = \frac{-4}{\sqrt{1-\frac{1}{4}}} = -\frac{8}{\sqrt{3}}$$

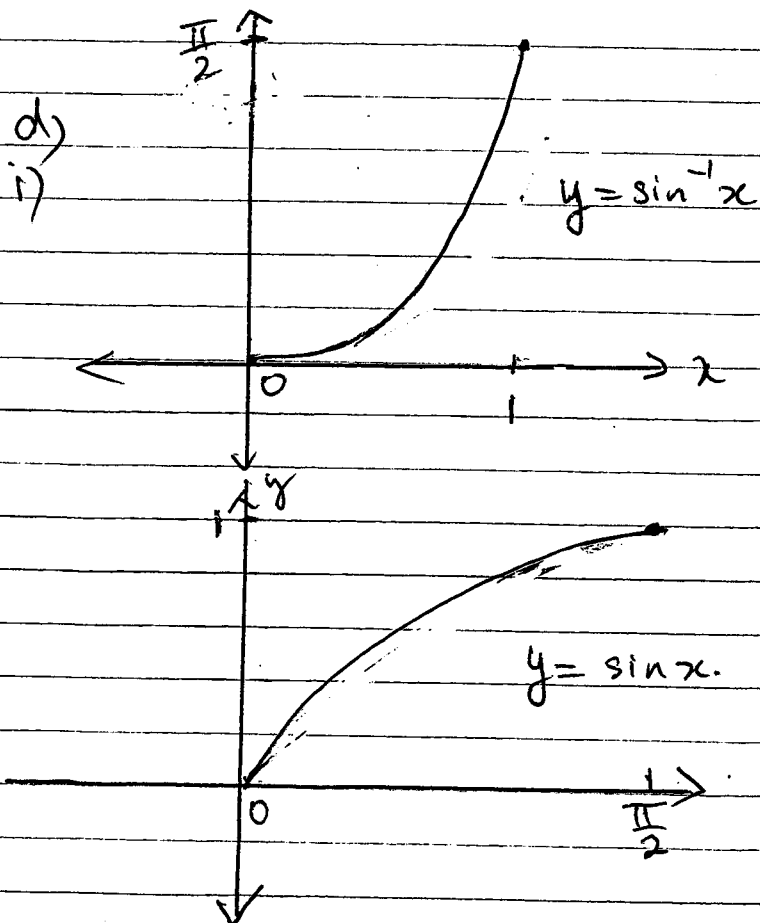
$$\text{at } x = -\frac{1}{8}, y = \cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$$

Equation of tangent

$$y - \frac{2\pi}{3} = -\frac{8}{\sqrt{3}} \left(x + \frac{1}{8}\right) \rightarrow y - \frac{2\pi}{3} = -\frac{8x}{\sqrt{3}} - \frac{1}{\sqrt{3}}$$

$$y = -\frac{8}{\sqrt{3}}x - \frac{1}{\sqrt{3}} + \frac{2\pi}{3} \rightarrow 24x + 3\sqrt{3}y + 3 - 2\sqrt{3}\pi$$

$$\begin{aligned} \text{ii) } \int_0^1 \sin^{-1} x \, dx + \int_0^{\frac{\pi}{2}} \sin x \, dx \\ = \frac{\pi}{2} \times 1 + 1 \\ = \frac{\pi}{2} + 1 \end{aligned}$$



Q7.

a) For $n=1$

$$4^1 + 14 = 18 \\ = 6 \times 3$$

$\therefore$ Statement is true for $n=1$.

Assume statement is true for $n=k$.

$$4^k + 14 = 6p, \text{ for } p \text{ some integer.} \quad 1$$

Prove that statement is true for $n=k+1$.

$$\begin{aligned} 4^{k+1} + 14 &= 4 \times 4^k + 14 \\ &= 4 \times 4^k + 4 \times 14 - 3 \times 14 \\ &= 4(4^k + 14) - 6 \times 7 \\ &= 4 \times 6p - 6 \times 7 \\ &= 6(4p - 7) \end{aligned} \quad 1$$

$\therefore 4^{k+1} + 14$ is divisible by 6.

Since statement is true for $n=1$ and assume true for $n=k$ and proved true for $n=k+1$, then it is true for $n=2, 3, 4, \dots$ i.e. all positive integral values of n .

b) (i) $f'(x) = \frac{1}{3} (x^3 - 12x)^{-\frac{2}{3}} (3x^2 - 12)$ or $(x^2 - 4)(x^3 - 12x)^{-\frac{2}{3}}$ 1

ii) $x_1 = 3.3$.

$$\begin{aligned} f(x_1) &= (-3.3^3 - 12 \times 3.3)^{\frac{1}{3}} \\ &\approx 1.54 \end{aligned}$$

$$\begin{aligned} f'(x_1) &= \frac{1}{3} (-3.3^3 - 12 \times 3.3)^{-\frac{2}{3}} (3 \times (-3.3)^2 - 12) \\ &\approx 2.90. \end{aligned}$$

$$\begin{aligned} x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= 3.3 - \frac{1.54}{2.90} \\ &= -3.83 \end{aligned}$$

(ii) Since graph at $x = -3.3$ is almost flat, then the tangent at $x = -3.3$ cuts the x axis further away from the actual x intercept (i.e. root of $f(x)$).

$$\begin{aligned}
 (c) T_{k+1} &= {}^9C_k (x^2)^{9-k} \left(\frac{-2}{x}\right)^k \\
 &= {}^9C_k x^{18-2k} (-2)^k x^{-k} \\
 &= {}^9C_k (-2)^k x^{18k-3k}
 \end{aligned}$$

Term independent of x is when
 $18-3k=0$

$$k=6$$

$$\begin{aligned}
 T_7 &= {}^9C_6 (-2)^6 x^0 \\
 &= 84 \times 64 \\
 &= 5376
 \end{aligned}$$

$$d) (1+x)^{n-1} = \binom{n-1}{0} 1^{n-1} + \binom{n-1}{1} 1^{n-2} x + \dots + \binom{n-1}{n-2} x^{n-2} + \binom{n-1}{n-1} x^{n-1}$$

(i) Let $x=1$

$$2^{n-1} = 1 + \binom{n-1}{1} + \dots + \binom{n-1}{n-2} + 1$$

$$\therefore 2^{n-1} - 2 = \binom{n-1}{1} + \binom{n-1}{2} + \dots + \binom{n-1}{n-2}$$

$$(ii) 2^{n-1} - 2 > 1000$$

$$2^{n-1} > 1002$$

$$(n-1) \ln 2 > \ln 1002$$

1

$$n-1 > \frac{\ln 1002}{\ln 2}$$

$$n > \frac{\ln 1002}{\ln 2} + 1$$

$$n > 10.9.$$

$\therefore$ least positive integer is $n=11$. 1